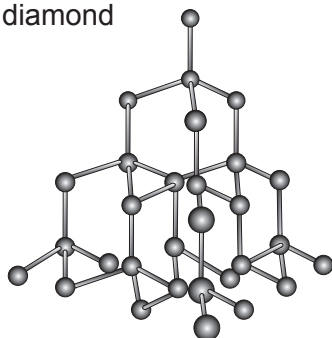


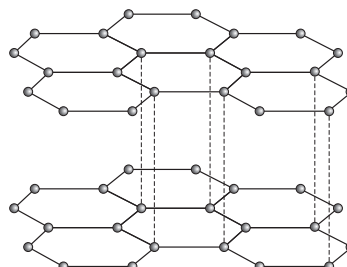
7. Allotropes of carbon include diamond, graphite, graphene, carbon fibres, fullerenes and carbon nanotubes.

(a) The structures of diamond and graphite are shown below.

diamond



graphite



Answer the following questions with reference to the bonding in diamond and graphite.

- (i) Explain why diamond is an insulator of electricity but graphite is a conductor. [4]

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- (ii) Explain why graphite is soft and slippery but diamond is hard. [3]

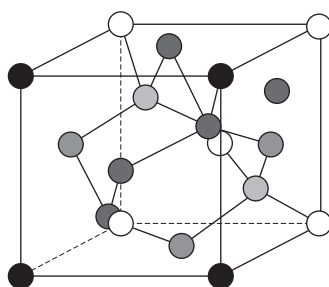
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(b) A unit cubic cell of diamond is shown below.



Each side has dimensions of 0.36 nm ($1 \text{ nm} = 10^{-9} \text{ m}$).

The number of carbon atoms in a cubic cell is 8.

The mass of a carbon atom is $2 \times 10^{-26} \text{ kg}$.

Use the information above and the equation:

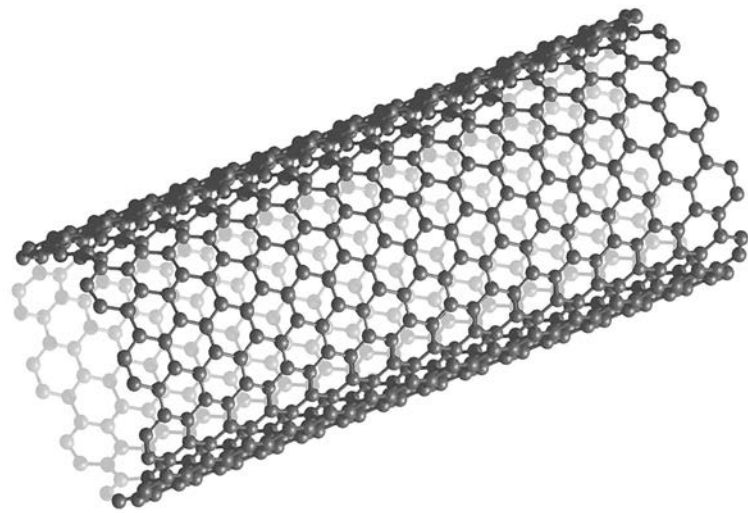
$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of this unit cubic cell of diamond to 2 significant figures.

[4]

Density = kg/m^3

(c) A carbon nanotube is shown below.



Some properties of these nanotubes are compared with three other materials in the table below.

Material	Tensile strength (MPa)	Density (g/cm ³)
stainless steel	505	8.00
aluminium	572	2.81
Kevlar	3 620	1.44
carbon nanotube	62 000	1.34

Explain why carbon nanotubes are likely to replace materials such as steel and aluminium in the manufacture of power lines. [2]

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END OF PAPER